Thermodynamic Equilibrium from Reversible One-Quantum Exchange among Discrete Particles

A finite-state counting argument and numerical realization

Matter-as-machine

Abstract

This note studies a minimal model of thermodynamic exchange among discrete particles. A particle is represented only by a nonempty finite list of quanta and is constrained to remain at size $n \geq 1$. Relative to that minimum size, it has excess occupancy $m = n - 1$. No particular list entry is a persistent baseline: an elementary event may transfer any selected quantum provided that the donor remains nonempty. Each transferred quantum carries energy ε, and total exchange energy is conserved. For M particles and Q excess quanta, the number of occupancy configurations is $\binom{Q+M-1}{Q}$. The uniform microcanonical measure therefore has an exact finite-system marginal distribution and, in the thermodynamic limit at fixed $\nu = \frac{Q}{M}$, a geometric marginal $P(m) = (1-r)r^m$ with $r = \frac{\nu}{1+\nu}$. Defining entropy by $S = k \log \Omega$ and temperature by $\frac{1}{T} = \frac{\partial S}{\partial E}$ gives $r = \exp\left[-\frac{\varepsilon}{kT}\right]$ and $\langle m \rangle = \frac{1}{\left[\exp\left(\frac{\varepsilon}{kT}\right)-1\right]}$. Thus the exponential thermal parameter is obtained from state counting rather than inserted as a change of variables. A reversible one-quantum Markov process is shown to have the uniform microcanonical measure as its unique stationary distribution. Simulations of two initially unequal reservoirs show energy equalization, vanishing late-time net flux, and approximately geometric particle-size statistics. In one reported simulation, a directional extension with six taxicab instructions and perpendicular-only exchange shows approximate agreement with the same equilibrium statistics while relaxing more slowly; no stationary-law theorem is claimed for that asymmetric process. The result is a minimal thermodynamics of a fixed-energy exchange ensemble, not a derivation of quantum mechanics, a full Bose gas, or the Planck spectrum.

1. Purpose and scope

The central question is deliberately narrow:

> Can temperature and a Bose-Einstein-shaped mean occupancy arise from a concrete one-quantum exchange process among discrete particles, without first imposing the substitution $r = \exp\left[-\frac{\varepsilon}{kT}\right]$?

The answer is yes for the idealized model defined below. The result has two parts:

- a counting result that determines the equilibrium distribution and its temperature parameter; and
- a stochastic dynamics that actually approaches that equilibrium by reversible one-quantum transfers.

> **Ontological postulate.** Matter is discrete. A particle is identified with a finite nonempty list of elementary instructions, not merely represented by such a list for mathematical convenience. The empty list represents absence of a particle. Consequently, an existing particle has total length $n \geq 1$. Writing $n = 1 + m$, the exchangeable occupancy satisfies $m \geq 0$. The condition $n \geq 1$ is a boundary on total list length; it does not label any particular instruction as a protected baseline.

This paper establishes consequences of that postulate only within the stated reversible exchange model. It does not establish that real elementary particles possess this structure.

Mathematically, the counting is the familiar distribution of indistinguishable quanta among distinguishable containers and is closely related to standard Bose-Einstein occupancy counting. At the level of stochastic occupancy dynamics, the model also overlaps with zero-range and related particle-transfer processes [6]. The claim is therefore not that the combinatorial formula or the general transfer-process mathematics is new. The distinct claim is interpretive: the occupancies are proposed as the literal discrete internal matter of particles, not as site occupancies in an auxiliary stochastic model. Scientific support for that ontology requires distinctive empirical predictions beyond reproducing established equilibrium statistics.

Several topics are intentionally excluded. There is no spatial escape in the equilibrium derivation, no cosmological loss or “tired light,” no creation of matter, no pressure-volume law, and no claim that the six-direction extension removes all taxicab anisotropy. These belong to separate models and should not be used to justify the equilibrium result proved here.

2. Discrete particles and conserved exchange energy

Consider $M \geq 2$ distinguishable particles. Particle i has total list length

$$n_i = 1 + m_i, \quad m_i \in \{0, 1, 2, \ldots\}.$$

The constraint $n_i \geq 1$ ensures that a particle always exists. The variable $m_i = n_i - 1$ is a count relative to this minimum, not a label attached to a protected instruction. Whenever $n_i > 1$, any selected instruction may be transferred while leaving at least one instruction behind. Each transferred instruction carries the same energy $\varepsilon > 0$. Hence

$$E_i = \varepsilon m_i, \quad E = \sum_{i=1}^{M} E_i = \varepsilon Q, \quad Q = \sum_{i=1}^{M} m_i.$$

An elementary exchange transfers one excess instruction:

$$(m_i, m_j) \to (m_i - 1, m_j + 1), \quad m_i \geq 1.$$

The reverse event is also allowed. Every event conserves Q and therefore E. No instruction is created or destroyed in this closed model.

The minimum-size subtraction is an energy convention, not a persistent physical quantum. If the total list energy is instead defined as εn_i, every particle receives the same additive offset ε. The total energy then changes by $M\varepsilon$, but exchange probabilities, entropy differences, temperature, and the distribution of m_i are unchanged.

3. Exact finite-system state counting

3.1. Number of configurations

A microconfiguration at the occupancy level is an ordered M-tuple

$$(m_1, \ldots, m_M), \quad m_i \geq 0, \quad \sum_i m_i = Q.$$

The number of such weak compositions is the stars-and-bars count

$$\Omega(Q, M) = \binom{Q + M - 1}{Q} = \binom{Q + M - 1}{M - 1}.$$

The microcanonical assumption is that all these occupancy configurations have equal stationary probability. Section 6 supplies an explicit Markov process for which this is not merely assumed but follows from detailed balance and ergodicity.

3.2. Exact marginal of one particle

Fix $m_1 = m$. The remaining $Q - m$ quanta may be distributed among $M - 1$ particles in

$$\Omega(Q - m, M - 1) = \binom{Q - m + M - 2}{Q - m}$$

ways. Therefore the exact finite-system marginal is

$$P_{Q,M}(m) = \frac{\binom{Q-m+M-2}{Q-m}}{\binom{Q+M-1}{Q}}, \quad 0 \leq m \leq Q.$$

Its adjacent ratio is

$$\frac{P_{Q,M}(m+1)}{P_{Q,M}(m)} = \frac{Q - m}{Q - m + M - 2}.$$

This ratio depends weakly on m at finite Q and M. Consequently the exact finite marginal is not exactly geometric. That distinction matters: the geometric law appears as a controlled thermodynamic-limit result, not as an identity for every finite ensemble.

3.3. Thermodynamic limit

Let $Q, M \to \infty$ while the mean excess occupancy

$$\nu = \frac{Q}{M}$$

remains fixed. For each fixed m,

$$\frac{Q-m}{Q-m+M-2} \to \frac{Q}{Q+M} = \frac{\nu}{1+\nu} = r.$$

Normalization then gives

$$P(m) = (1-r)r^m, \quad m \geq 0, \quad r = \frac{\nu}{1+\nu}.$$

The mean of this distribution is

$$\langle m \rangle = \frac{r}{1-r} = \nu.$$

Thus the geometric law is both the limiting marginal of uniform compositions and the distribution whose mean equals the conserved excess occupancy per particle.

4. Entropy and the emergence of temperature

Define microcanonical entropy by

$$S(E, M) = k \log \Omega(Q, M), \quad Q = \frac{E}{\varepsilon},$$

where k fixes the temperature unit. Using Stirling's approximation at large Q and M gives

$$\frac{S}{k} \approx (Q+M)\log(Q+M) - Q\log Q - M\log M.$$

In terms of $\nu = \frac{Q}{M}$,

$$S \approx kM[(1+\nu)\log(1+\nu) - \nu\log\nu].$$

Temperature is defined thermodynamically by

$$\frac{1}{T} = \frac{\partial S}{(\partial E)_M}.$$

Applying this definition to the Stirling-limit entropy, and using $E = \varepsilon Q$, gives in the thermodynamic limit

$$\frac{1}{T} \approx \left(\frac{k}{\varepsilon}\right)\log\left(\frac{Q+M}{Q}\right) = \left(\frac{k}{\varepsilon}\right)\log\left(1+\frac{1}{\nu}\right).$$

Equivalently, in that same limit,

$$\frac{\varepsilon}{kT} \approx \log\left(1+\frac{1}{\nu}\right).$$

Solving the limiting relation for ν yields

$$\nu = \langle m \rangle = \frac{1}{\exp\left(\frac{\varepsilon}{kT}\right) - 1}.$$

The limiting geometric ratio becomes

$$r = \frac{\nu}{1+\nu} = \exp\left[-\frac{\varepsilon}{kT}\right].$$

The full limiting marginal is therefore

$$P(m) = \left(1 - \exp\left[-\frac{\varepsilon}{kT}\right]\right)\exp\left[-m\frac{\varepsilon}{kT}\right].$$

> The exponential factor has not been postulated. It follows from the logarithmic growth of the number of configurations and the thermodynamic definition
>
> $$\frac{1}{T} = \frac{\partial S}{\partial E}.$$

This is the mathematical single-energy Bose-Einstein occupancy form. It is not yet the full distribution of radiation over frequency. A Planck spectrum also requires an energy label such as $\varepsilon = \hbar\omega$, a density of states, and the appropriate photon interpretation.

5. Thermodynamic relations of the ideal exchange ensemble

The relations in this section use the thermodynamic-limit temperature obtained in Section 4. They are not asserted as exact finite-M, Q identities.

5.1. Internal energy

The mean exchange energy per particle is

$$u(T) = \varepsilon\langle m\rangle = \frac{\varepsilon}{\exp(\frac{\varepsilon}{kT}) - 1}.$$

For M particles,

$$U(T, M) = M\frac{\varepsilon}{\exp(\frac{\varepsilon}{kT}) - 1}.$$

Any baseline energy is additive and omitted here.

5.2. Heat capacity

Let $x = \frac{\varepsilon}{kT}$. Differentiation gives the heat capacity per particle

$$c(T) = \frac{\mathrm{d}u}{\mathrm{d}T} = kx^2\frac{\exp(x)}{(\exp(x) - 1)^2} > 0.$$

At high temperature, $x \ll 1$ and

$$u(T) \approx kT - \frac{\varepsilon}{2} + O(T^{-1}), \quad c(T) \to k.$$

At low temperature,

$$u(T) \approx \varepsilon \exp\left[-\frac{\varepsilon}{kT}\right], \quad c(T) \to 0.$$

5.3. Zeroth law: an operational temperature

Place two subsystems A and B in weak exchange contact while conserving $E_A + E_B$. The total multiplicity is

$$\Omega_{\text{tot}(E_A)} = \Omega_{A(E_A)}\Omega_{B(E-E_A)},$$

and the most probable division maximizes $S_{\text{tot}} = S_A + S_B$. At the maximum,

$$\frac{\partial S_A}{\partial E_A} = \frac{\partial S_B}{\partial E_B},$$

so

$$\frac{1}{T_A} = \frac{1}{T_B}.$$

Operationally, two bodies have the same temperature when their late-time mean energy flux vanishes. This is stronger than identifying temperature with an interaction frequency: changing the event rate changes how quickly equilibrium is reached, whereas changing $\frac{Q}{M}$ changes which equilibrium is reached.

5.4. First law in the closed exchange model

Every transfer removes one quantum from one particle and gives it to another. Therefore

$$\Delta E_A + \Delta E_B = 0.$$

There is heat exchange but no work term in the present model. Its first-law content is simply conservation of exchange energy.

5.5. Second law as a typicality statement

Starting far from equilibrium, overwhelmingly more configurations correspond to energy divisions near the maximum of $S_A + S_B$ than to extreme divisions. A reversible stochastic process therefore moves toward the high-multiplicity region with high probability. Microscopic trajectories can fluctuate away from it; the statistical claim is that the equilibrium macrostate dominates for large systems.

The model also has a simple low-temperature limit. As $T \to 0$, $\nu \to 0$, all excess occupancies vanish, and the unique occupancy configuration is $m_i = 0$ for every particle. Hence $S \to 0$ within this idealized occupancy description.

6. Reversible one-quantum exchange dynamics

The counting above describes equilibrium. We now give a process that reaches it. At each discrete update:

1. Choose an ordered pair of distinct particles (i, j) uniformly.

2. If $m_i > 0$, transfer one quantum from i to j.
3. If $m_i = 0$, perform no transfer.

For any two neighboring configurations a and b connected by moving one quantum from i to j,

$$P(a \to b) = \frac{1}{[M(M-1)]} = P(b \to a).$$

Thus the transition matrix is symmetric on every allowed edge. The uniform measure

$$\pi(a) = \frac{1}{\Omega(Q, M)}$$

satisfies detailed balance:

$$\pi(a)P(a \to b) = \pi(b)P(b \to a).$$

The state graph is connected because quanta can be transferred one at a time between any particles. The self-loops at empty donors make the chain aperiodic. The finite chain is therefore irreducible and aperiodic, so the uniform microcanonical distribution is its unique stationary law and the process converges to it from every initial configuration.

This argument identifies the essential assumptions:

- one-quantum transfers;
- conservation of Q;
- reversibility with equal forward and reverse edge weights; and
- sufficient connectivity to explore the full fixed-Q state space.

If physical exchange rules violate these conditions, a different stationary law may result.

7. Two-reservoir numerical experiment

7.1. Nondirectional control model

The implementation `two_bath_reversible_exchange.cpp` used two reservoirs with 1000 particles each. Reservoir A began with mean excess occupancy 8 and reservoir B with mean excess occupancy 1. Thus the conserved global mean was

$$\nu_{\text{global}} = \frac{8+1}{2} = 4.5.$$

After 50 million attempted transfers, representative late-time values were

stage	ν_A	ν_B	net cross-flux
initial	8.000	1.000	positive
late representative state	4.477	4.523	-2.0×10^{-5}
equilibrium target	4.500	4.500	0

The equilibrium parameters predicted from the conserved global mean are

$$r = \frac{4.5}{5.5} = 0.818182, \quad \beta\varepsilon = \log\left(1 + \frac{1}{4.5}\right) = 0.200671.$$

The reservoir means fluctuate around 4.5 rather than remaining exactly equal at every snapshot, as expected for finite systems.

7.2. Directional particles and perpendicular exchange

For a directional extension, every list entry belongs to

$$\mathcal{D} = \{R, L, U, D, F, B\}.$$

Each particle selects an active instruction uniformly from its list. Two active instructions are compatible only when their coordinate axes are perpendicular. For an isotropic six-direction mixture, the compatible fraction is

$$P_{\text{perp}} = \frac{4}{6} = \frac{2}{3}.$$

When a pair is compatible, either particle is chosen as donor with probability $\frac{1}{2}$; if the selected donor has excess occupancy, its active instruction is transferred to the receiver. The total number of instructions is conserved.

The implementation `two_bath_perpendicular_exchange.cpp` used 2000 particles per reservoir, the same initial means 8 and 1, a 20% probability of a cross-reservoir encounter, and 200 million attempted interactions. It measured

quantity	perpendicular only	all directions
late ν_A	4.465	4.693
late ν_B	4.535	4.308
successful transfers per cross attempt	0.546	0.818
late net flux per cross attempt	4.1×10^{-5}	-1.26×10^{-4}
geometric TV distance, A	0.039	0.049
geometric TV distance, B	0.035	0.033

The unequal late values in a single snapshot are finite-size fluctuations. The time series oscillates around the common target 4.5, and the late net flux oscillates around zero.

The ratio of transfer frequencies is

$$\frac{0.546}{0.818} \approx 0.667,$$

matching the expected perpendicular compatibility fraction $\frac{2}{3}$. The perpendicular rule therefore changes the relaxation rate but does not appear to create the thermal equilibrium law. This control is important: the geometric equilibrium is attributable to reversible conserved exchange, while perpendicularity remains a separate kinematic restriction.

7.3. Why the simulated TV distance is not zero

There are three sources of visible discrepancy from a perfect geometric curve:

- the exact finite-system marginal is not exactly geometric;
- a histogram from a finite snapshot has sampling noise; and
- selecting active instructions proportionally to their multiplicities changes microscopic transition weights in the directional implementation, so its occupancy projection is not identical to the ideal symmetric-edge chain.

The numerical result should therefore be described as approximate agreement in the reported simulation, not as an exact computational proof or evidence of a proved directional stationary law.

8. Relation to Bose-Einstein statistics

The limiting marginal

$$P(m) = (1 - \exp[-\beta\varepsilon]) \exp[-\beta\varepsilon m]$$

and mean

$$\langle m \rangle = \frac{1}{\exp(\beta\varepsilon) - 1}$$

are the standard single-energy bosonic occupancy formulas at zero chemical potential. Bose's original radiation counting and Einstein's extension to an ideal gas established the historical quantum-statistical setting for these expressions [1, 2].

The present construction differs in interpretation. The distinguishable containers are taken to be discrete particles represented by lists, and the quanta are transferable list entries. This interpretation is an additional physical hypothesis; the combinatorics alone does not establish that actual particles have this structure.

Three distinctions prevent overstatement:

1. **Single energy versus a spectrum.** This note assumes one quantum energy ε. A Planck spectrum requires a family of energies and a density of states.
2. **Occupancy mathematics versus quantum mechanics.** The geometric law does not derive wave functions, commutation relations, indistinguishability from first principles, or bosonic exchange symmetry.
3. **Equilibrium versus formation and escape.** The closed microcanonical model does not describe photons leaving a source region. Open-system escape is a different stochastic problem.

9. What has and has not been derived

Within the stated model, the following results are derived:

- the exact finite marginal $P_{Q,M}(m)$;
- its geometric thermodynamic limit;
- entropy $S = k \log \Omega$;
- the thermodynamic-limit temperature relation $r = \exp\left[-\frac{\varepsilon}{kT}\right]$;
- the Bose-Einstein-shaped mean occupancy;
- internal energy and heat capacity;
- equality of temperature as the condition for maximal total entropy and zero late-time mean flux; and
- convergence of a specified reversible exchange chain to the microcanonical equilibrium.

The following are not derived:

- the microscopic origin or numerical value of ε;
- the physical identity of the transferable quanta;
- the final directional exchange law of a matter-as-machine theory;
- the Planck spectral factor or density of states;
- pressure, volume, mechanical work, or an equation of state;
- chemical potential when particle number changes;
- creation, annihilation, reset events, or cosmological redshift; and
- quantum mechanics or the ontology of real elementary particles.

This boundary is not cosmetic. It separates a reproducible mathematical result from hypotheses that require additional dynamics and empirical tests.

10. Reproducibility

The simulations are deterministic given their pseudorandom seeds and are implemented in the accompanying source files:

- `two_bath_reversible_exchange.cpp`: symmetric nondirectional exchange;
- `two_bath_perpendicular_exchange.cpp`: directional and all-direction controls;
- `test_geometric_robustness.cpp`: geometric-fit robustness in the separate open escape model; and
- `predict_endpoints_from_escape_chain.cpp`: a coarse-grained prediction test for that open model.

Only the first two are evidence for the closed equilibrium claims of this note. The latter two are listed to make the separation from the open formation model explicit.

The reproducibility archive is the public repository github.com/matterasmachine/matterasmachine. The simulation code used for the reported runs is fixed at commit `6cf453728bbf76d8ad3ef2f1c5755c42be30528b`. The corresponding project folder can be viewed directly at this permanent revision. Later manuscript edits may continue to cite this commit as long as the simulation code and reported numerical results remain unchanged.

For each reported equilibrium run, one should retain the complete time series, repeat several independent seeds, and report confidence intervals rather than a single late snapshot. The present numbers are validation runs, not precision estimates of universal constants.

11. Conclusion

A minimal reversible exchange process among nonempty discrete particles is enough to produce a coherent ideal thermodynamics. Uniform fixed-energy configurations have an exact finite marginal that approaches a geometric law. The logarithm of their multiplicity supplies entropy; differentiating entropy with respect to energy supplies temperature; and the resulting thermodynamic-limit geometric ratio is $r = \exp\left[-\frac{\varepsilon}{kT}\right]$. A symmetric one-quantum Markov process converges to this equilibrium, while two-reservoir simulations show energy equalization and vanishing late-time net flux.

In the reported run, the six-direction perpendicular-exchange extension showed approximate agreement with the qualitative equilibrium statistics and reduced the interaction rate by the expected factor $\frac{2}{3}$. Because its transition weights are state-dependent and asymmetric, this observation is numerical only; it does not establish that the directional process has the same stationary law. It supports treating perpendicularity as a possible kinematic restriction, but not as the proved origin of the thermal law.

The main conceptual result is therefore modest but concrete:

For discrete particles that reversibly exchange conserved equal-energy quanta, temperature measures the equilibrium energy-per-particle scale selected by the number of accessible configurations. The Bose-Einstein denominator is the resulting occupancy relation, not an independently imposed exponential rule.

12. References

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Appendix A: finite marginal and geometric convergence

For completeness, expand the exact marginal as a product. From the adjacent ratio,

$$P_{Q,M}(m) = P_{Q,M}(0) \prod_{j=0}^{m-1} \frac{Q-j}{Q-j+M-2}.$$

Also,

$$P_{Q,M}(0) = \frac{M-1}{Q+M-1}.$$

At fixed $\nu = \frac{Q}{M}$ and fixed m,

$$P_{Q,M}(0) \to \frac{1}{1+\nu} = 1-r$$

and every factor in the finite product approaches

$$\frac{Q}{Q+M} = \frac{\nu}{1+\nu} = r.$$

Therefore

$$P_{Q,M}(m) \to (1-r)r^m.$$

The restriction to fixed m is the ordinary pointwise thermodynamic-limit statement. Uniform error bounds over a range of m require specifying how that range grows with M.

Appendix B: a finite-difference temperature

Before taking derivatives, define a discrete inverse temperature from the entropy gained when one quantum is added. The exact binomial ratio is $\frac{\Omega(Q+1,M)}{\Omega(Q,M)} = \frac{Q+M}{Q+1}$, and hence

$$\frac{\Delta S}{\varepsilon} = \left(\frac{k}{\varepsilon}\right) \log\left(\frac{Q+M}{Q+1}\right) \to \frac{1}{T} = \left(\frac{k}{\varepsilon}\right) \log\left(\frac{Q+M}{Q}\right).$$

The arrow denotes the large-Q, M limit. This formula makes clear that temperature measures the multiplicative increase in accessible configurations per added energy quantum.